

Supplementary Information for:
‘Detecting sparse biological invasions through multi-scale remote sensing: a drone–satellite framework for mapping *Neltuma* in dryland ecosystems’

Table S1. Details of the five spectral bands sampled by the DJI Phantom 4 Multispectral sensor array (the 6th camera is a standard RGB camera).

	Name	Central Wavelength (nm)	Band width (nm)
Band 1	Blue	450	32
Band 2	Green	560	32
Band 3	Red	650	32
Band 4	Red-edge	730	32
Band 5	Near-infrared	840	32

Table S2. Details of multispectral drone surveys flown for each study area.

AOI ID	Date	Total photos	Time Start	Average Wind Speed m/s	Total area (m²)	Spatial grain /ground surface distance (cm)
Bokspit_1	10/09/2022	3165	12.18	3.2	23,0340	5.8
Bokspits_2	12/09/2022	6060	12.15	2.4	43,1781	6.0
Bokspits_3	16/09/2022	4970	12.10	3.1	36,2680	6.1
Struizendam_1	18/09/2022	5645	11.34	4.0	42,0433	5.9
Struizendam_2	18/09/2022	4810	12.39	1.9	35,7212	6.7
Struizendam_3	20/09/2022	4780	11.25	1.2	34,2298	6.5
Struizendam_4	21/09/2022	2370	13.35	3.9	18,1654	5.9

Table S3. A definition of the vegetation/land cover types used in the study.

Class	Description
Bare Ground	Non vegetated sand/soil/calcrete. Generally red in colour sand but includes grey /white calcrete ridges and mixed sand and calcrete
Grass	The grass species were senescent at the time of the field survey (dry season), dry and grey in colour. Heavily grazed near the villages. Grass assemblage was dominated by <i>Schmidtia kalahariensis</i> with <i>S. uniplumis</i> , <i>Eragrostis lehmanniana</i> and <i>Aristida meridionalis</i>
<i>Stipagrostis amabilis</i>	Commonly known as Kalahari Dune Grass this is a distinct and robust large grass forming bands of vegetation on dune crests in isolated locations on the study area
<i>Neltuma</i>	Formerly known as <i>Prosopis</i> and locally known as Mesquite or Sexanana. Mostly still in green leaf during study. Widely distributed with some dense patches in villages. Small to medium sized bushes in most of the study area. Larger established trees in villages.
<i>Vachellia erioloba</i>	Formerly known as <i>Acacia erioloba</i> and commonly known as Camel Thorn this tree is an important shade tree. Mostly occurring as large trees in the study area with rare smaller trees and bushes.
<i>Senegalia mellifera</i>	Formerly known as <i>Acacia mellifera</i> widespread across the study area occurring as small and medium sized bushes with some larger trees.
<i>Boscia albitrunca</i>	Commonly known as Shepherds Tree occurs in low numbers as isolated large individual trees across the study area. Evergreen with small dark green leaves.
Mixed woody cover	Mixture of woody vegetation dominated by <i>Rhigozum trichotomum</i> and <i>Senegalia mellifera</i> with <i>Vachellia haematoxylon</i> , <i>Vachellia erioloba</i> . Used as a vegetation class in satellite data classification where pixels not dominated by one tree species and did not include <i>Neltuma</i> .
<i>Rhigozum trichotomum</i>	Perennial woody shrub widespread across the study area growing 1-2 m tall. Forms dense thickets in places especially in overgrazed areas close to settlements.
<i>Gnidia polycephala</i>	Small low shrub forming isolated patches in one of the drone survey areas. Flowering with bright yellow flowers at the time of the survey. NB only marginal cover in Bospits_1 survey area
<i>Vachellia hebeclada</i>	Commonly known as Candle Thorn, this low bush grows often in circular patches in over grazed areas near settlements. NB only marginal cover in Bospits_1 survey area

Table S4. Number of training observations used for machine learning classification by land-cover class and drone survey area.

Land-cover class	Bokspits 1	Bokspits 2	Bokspits 3	Struizendam 1	Struizendam 2	Struizendam 3	Struizendam 4	Total
<i>Neltuma</i>	38	30	38	29	29	20	30	214
Bare Ground	30	24	33	24	31	30	30	202
Grass	30	30	28	37	30	30	0	185
<i>Gnidia polycephala</i>	20	20	21	0	0	0	0	61
<i>Vachellia erioloba</i>	2	28	27	35	19	0	2	113
<i>Rhigozum trichotomum</i>	20	16	57	11	37	30	20	191
<i>Senegalia mellifera</i>	0	16	18	0	0	30	0	64
<i>Boscia albitrunca</i>	0	0	0	0	0	14	0	14
<i>Stipagrostis amabilis</i>	0	10	0	0	0	0	0	10
Total	140	174	222	136	146	124	82	1,024

Table S5. Information about satellite data.

Parameter	WorldView 2	PlanetScope Super Dove multispectral	Sentinel-2
Date and time of acquisition	2021-10-19 08:48	2022-09-07 08:23	2022-09-07
Number of bands	4	4	4
Collection / product name	M2AS Surface Reflectance	PSScene 3B Surface Reflectance	Surface Reflectance (L2A)
Spatial resolution	1.6 m and 1.8 m	3 m	10 m
Cloud cover	None	None	None

Table S6. Equations for calculating vegetation indices from calibrated reflectance orthomosaics and satellite images.

Vegetation Indices	Formulae	Reference
Normalised Difference Vegetation Index (NDVI)	$\frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + \rho_{RED}}$	(Rouse et al. 1974)
Soil Adjusted Vegetation Index (SAVI)	$\frac{(1 + L) * (\rho_{NIR} - \rho_{RED})}{(\rho_{NIR} + \rho_{RED} + L)}$	(Huete 1988)
Modified Soil Adjusted Vegetation Index (MSAVI2)	L=0.5 $\frac{2 * \rho_{NIR} + 1 - \sqrt{((2 * \rho_{NIR} + 1)^2 - 8 * (\rho_{NIR} - \rho_{RED}))}}{2}$	(Qi et al. 1994)
Modified triangular vegetation index (MTVI2)	$\frac{1.5 * (1.2 * (\rho_{NIR} - \rho_{Green}) - 2.5 * (\rho_{RED} - \rho_{Green}))}{\sqrt{((2 * \rho_{NIR} + 1)^2 - (6 * \rho_{NIR} - 5 * \sqrt{\rho_{RED} - 0.5}))}}$	(Haboudane et al. 2004)

Table S7. Selection criteria for representative training pixels for each vegetation type based on fractional cover.

Criteria	WV2 1.6 m resolution pixels	PlanetScope 3 m resolution pixels	Sentinel-2 10 m resolution pixels
Minimum pixel fractional cover of vegetation type extracted from drone image classification	95%	85%	65%
Class size – number of training pixels selected per classification class	280	200	100

Table S8. Percentage cover of *Neltuma* used for definition of invasion phases.

cover (after Müllerová et al. 2023).

Invasion Phase	Percentage Cover of 250 m hexagon grid cell
Dominance	> 15%
Expansion	1.5 - 15%
Initial Incursion	0.1 – 1.5%
Pre-Incursion	< 0.1%

Table S9. High classification accuracy was obtained at the plant scale across the full range of *Neltuma* growth forms and phenological states. For every *Neltuma* plant occurring in the intensive drone survey areas (n = 184) predictions were obtained by extracting the majority classification from a 20 cm radius circle located in each plant's centre. The process was repeated for other plants and a representative selection of grass and bare ground.

Predicted by classification of drone data	Observed via ground-based survey						
		<i>Neltuma</i>	Bare Ground	Grass	Vachellia erioloba	Other woody	Total
	<i>Neltuma</i>	156	1	0	3	4	164
	Bare Ground	1	100	12	0	1	114
	Grass	4	2	97	2	9	114
	Vachellia Erioloba	7	1	1	38	9	56
	Other woody	16	1	1	17	43	78
	Total	184	105	111	60	66	526

Table S10. Confusion table evaluating classification of WV2 performance relative to classification of drone data.

Predicted by classification of WorldView 2 data	Predicted by classification of drone data						
		<i>Neltuma</i>	Bare Ground	Grass	Vachellia erioloba	Other woody	Total
	<i>Neltuma</i>	15371	8111	2689	2308	3577	32056
	Bare Ground	1603	185018	25528	1220	2049	215418
	Grass	2149	64836	214047	7205	21912	310149
	Vachellia Erioloba	3202	1000	513	2583	1703	9001
	Other woody	2131	7242	8337	3992	51051	72753
	Total	24456	266207	251114	17308	80292	32056

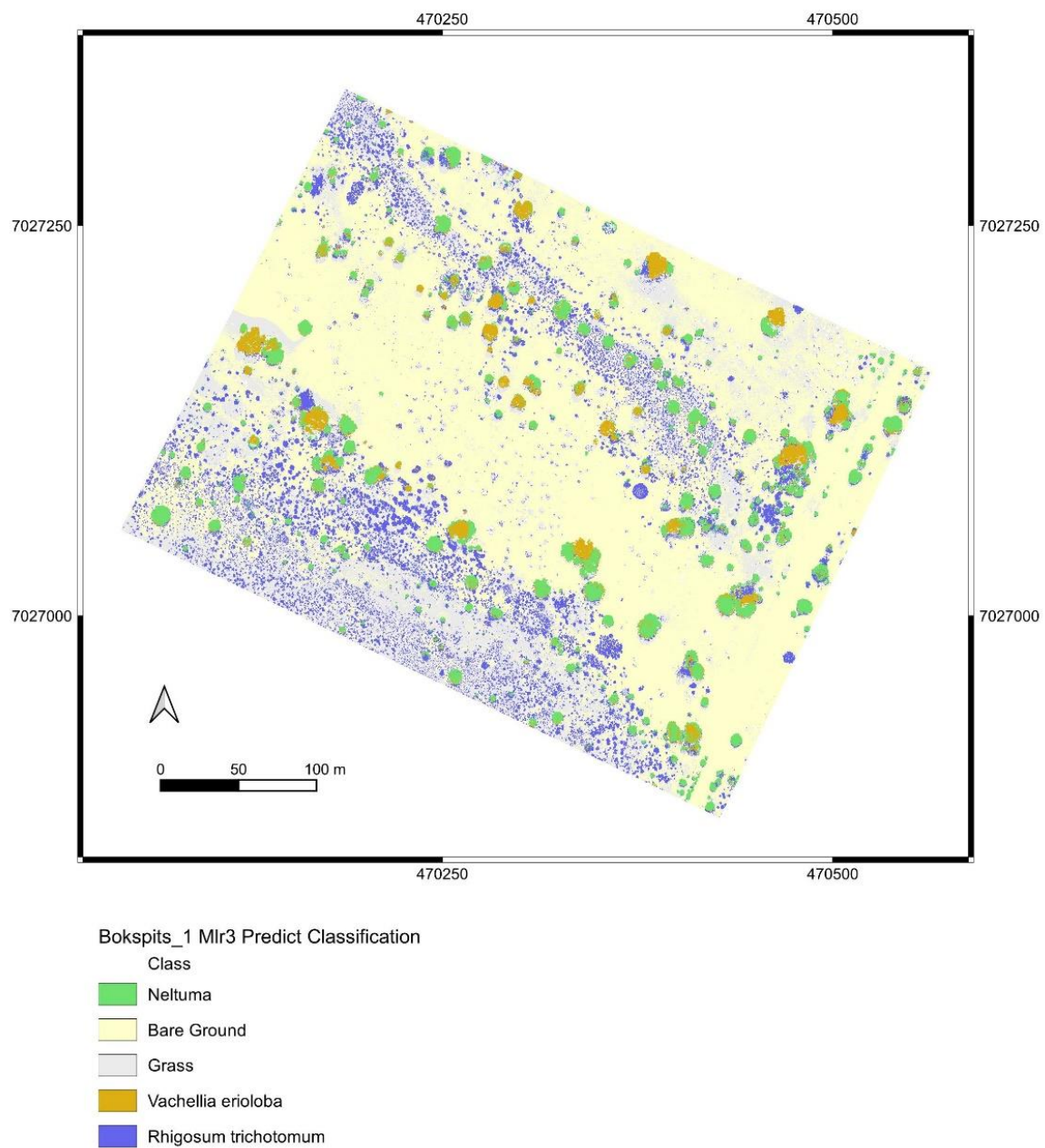


Figure S1. Final classification maps produced for the “Bokspits 1” drone survey area using the mlr3 pipeline. Best performing classification using SVM learner and Drone+CHM+AllVI data set.

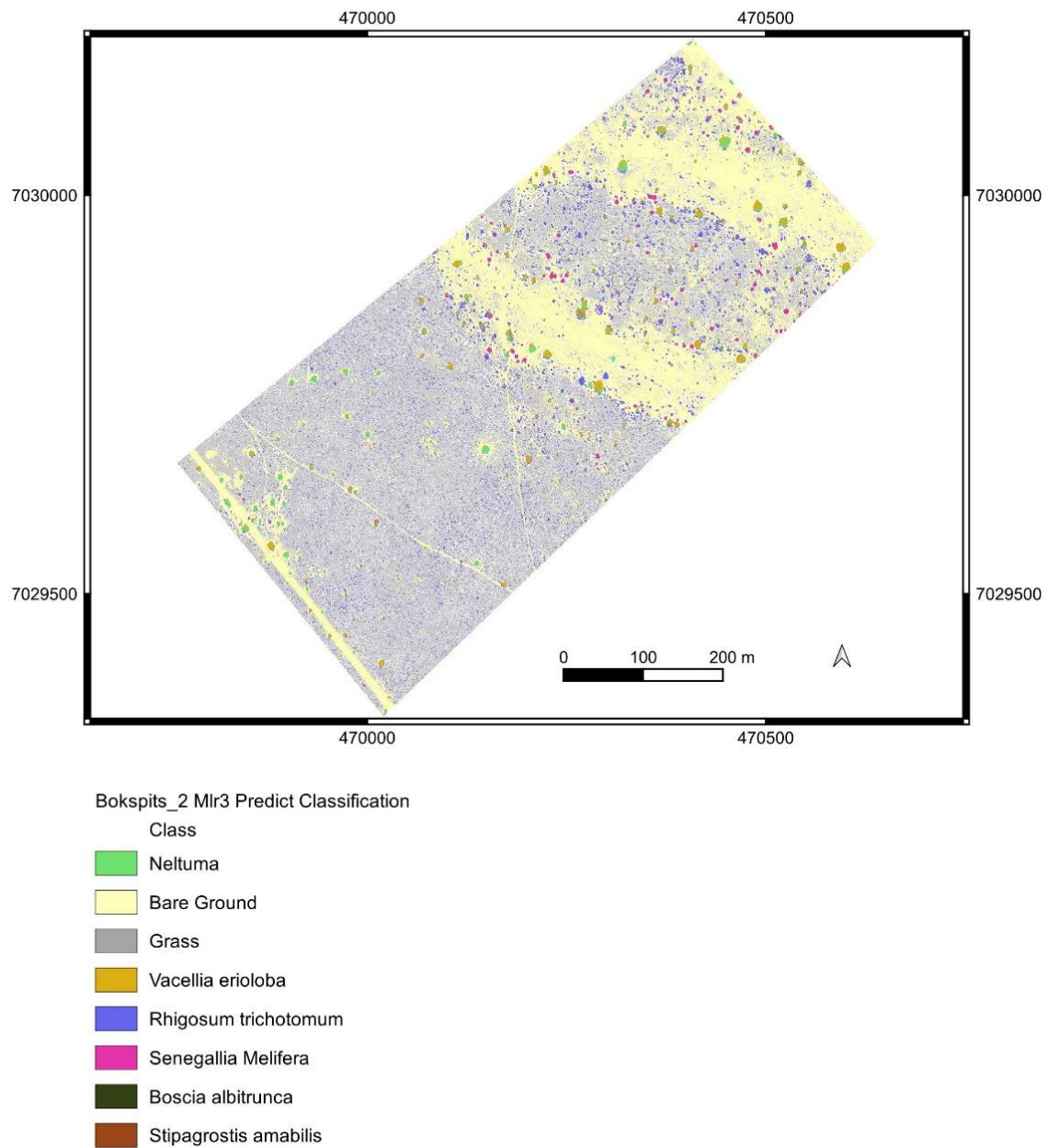


Figure S2. Final classification maps produced for the “Bokspits 2” drone survey area using the mlr3 pipeline. Best performing classification using SVM learner and Drone+CHM+AllVI data set.

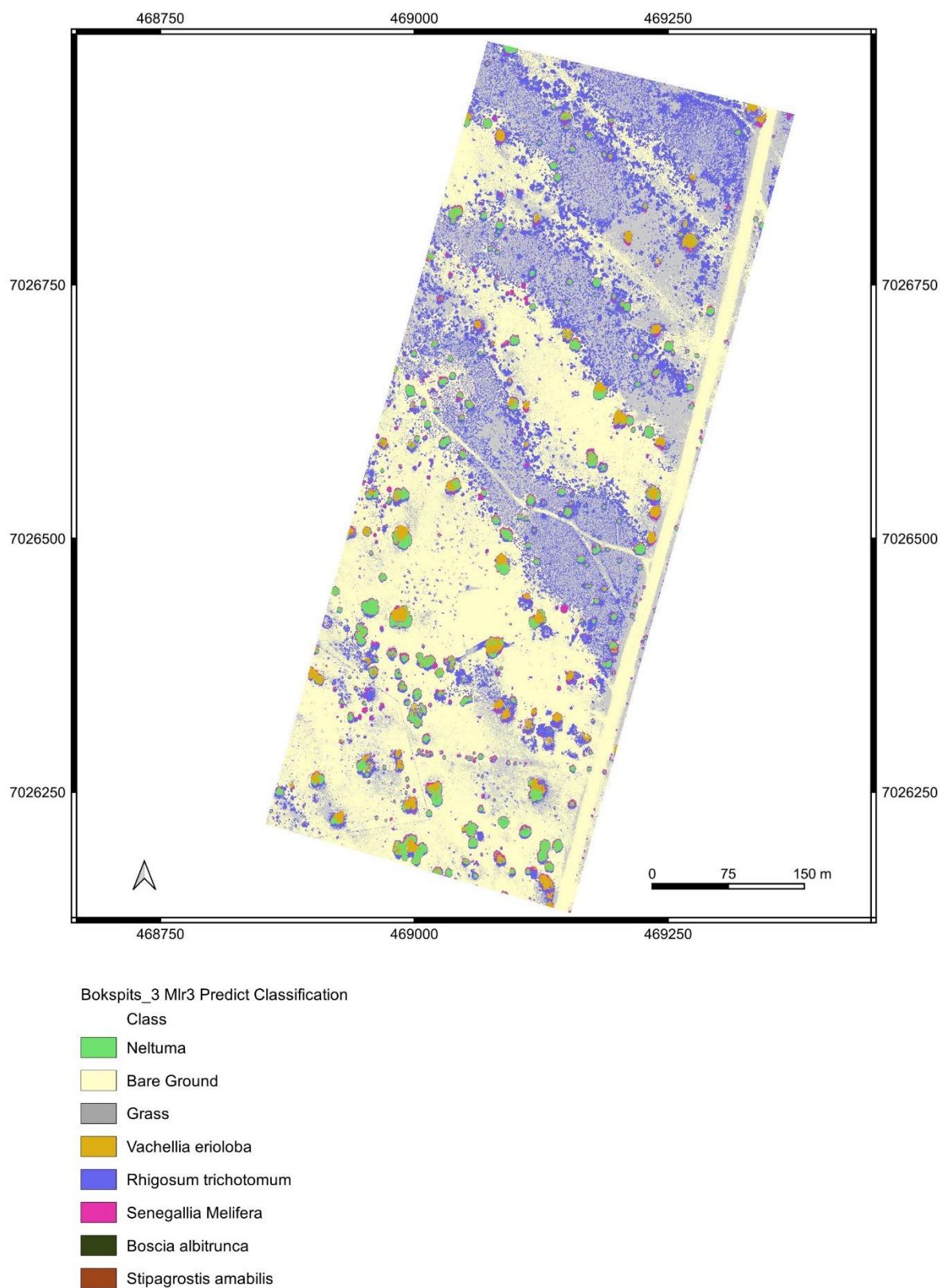


Figure S3. Final classification maps produced for the “Bokspits 3” drone survey area using the mlr3 pipeline. Best performing classification using SVM learner and Drone+CHM+AllVI data set.

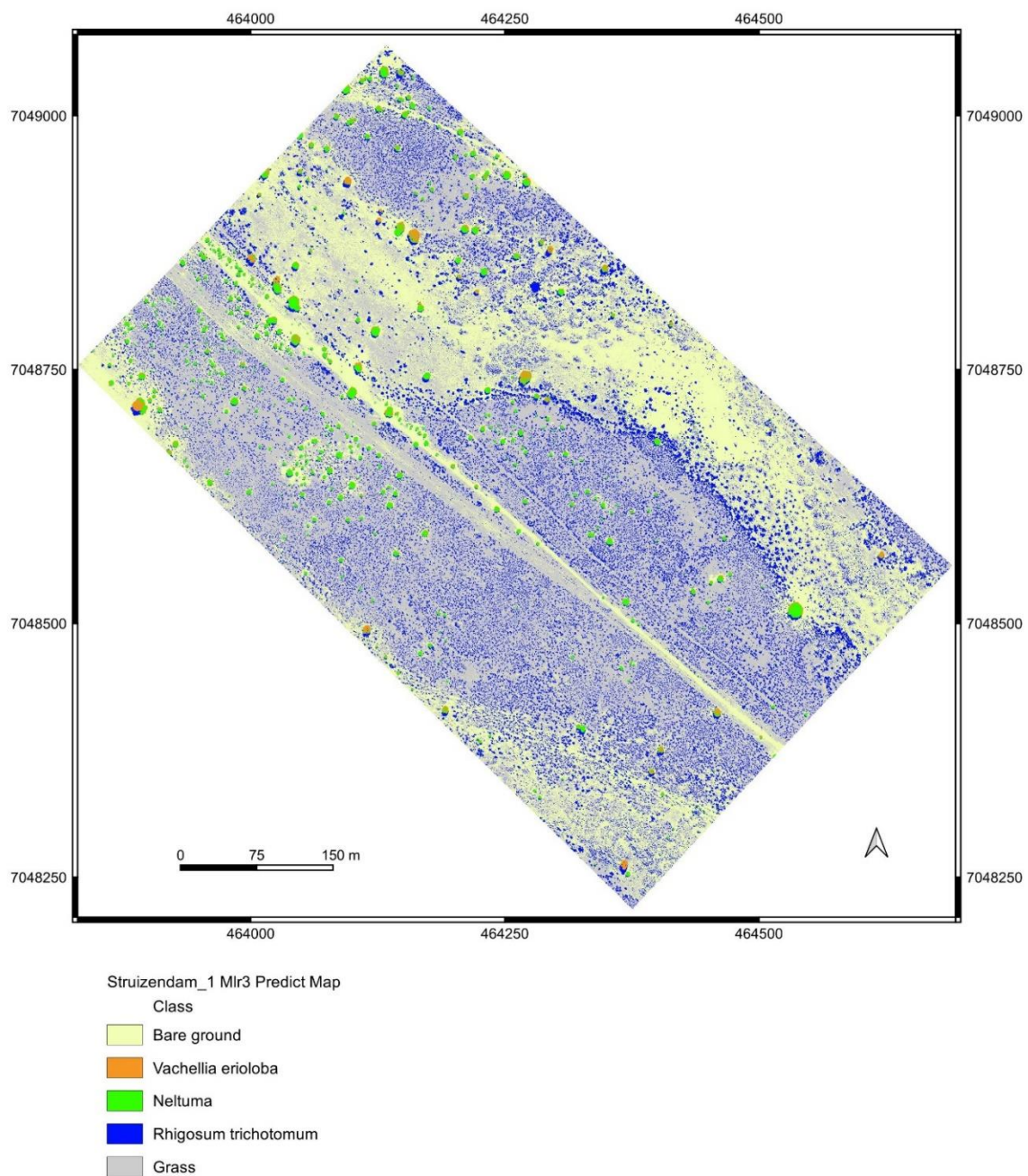


Figure S4. Final classification maps produced for the “Struizendam 1” drone survey area using the mlr3 pipeline. Best performing classification using SVM learner and Drone+CHM+AIIVI data set.

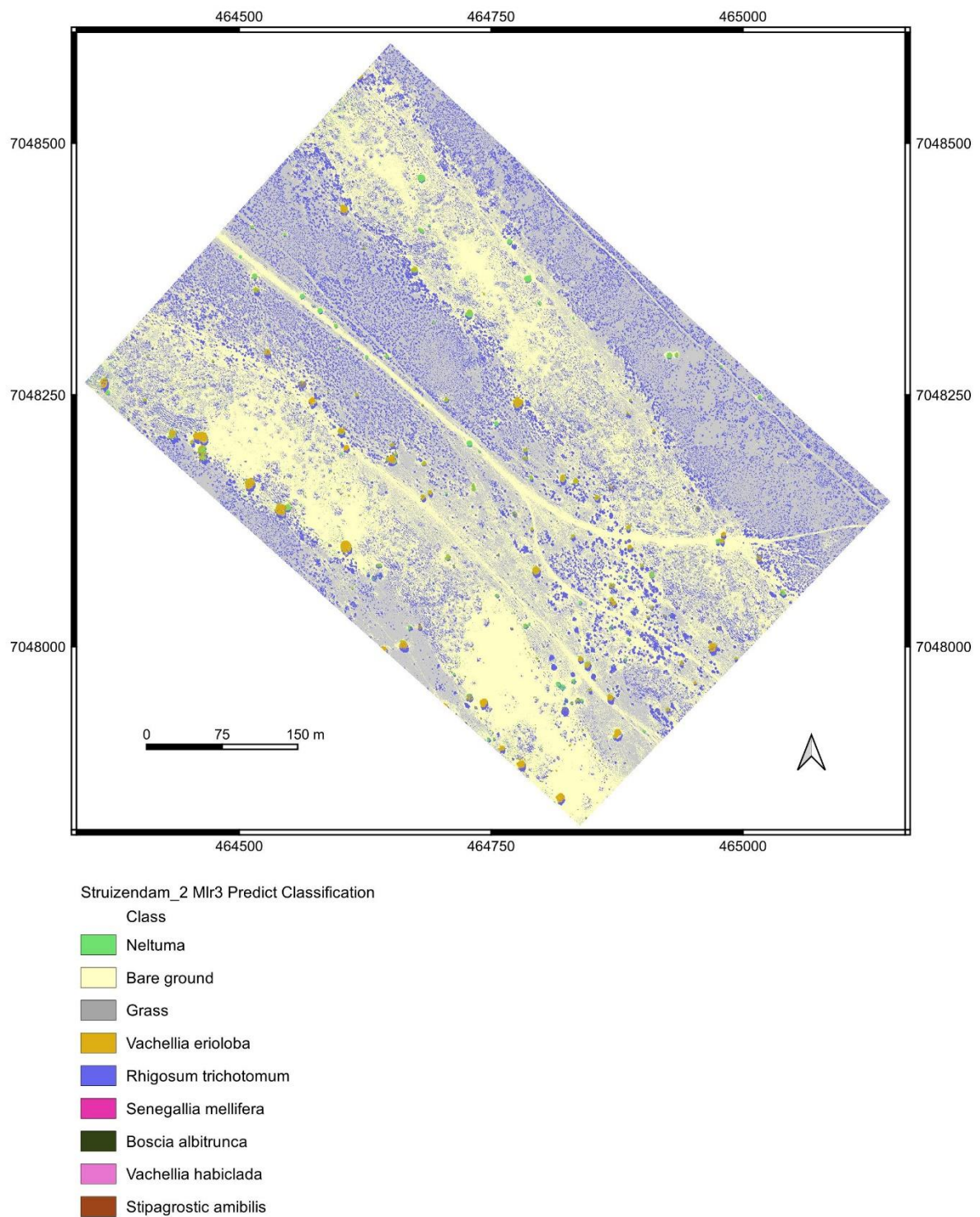


Figure S5. Final classification maps produced for the “Struizendam 2” drone survey area using the mlr3 pipeline. Best performing classification using SVM learner and Drone+CHM+AIIVI data set.

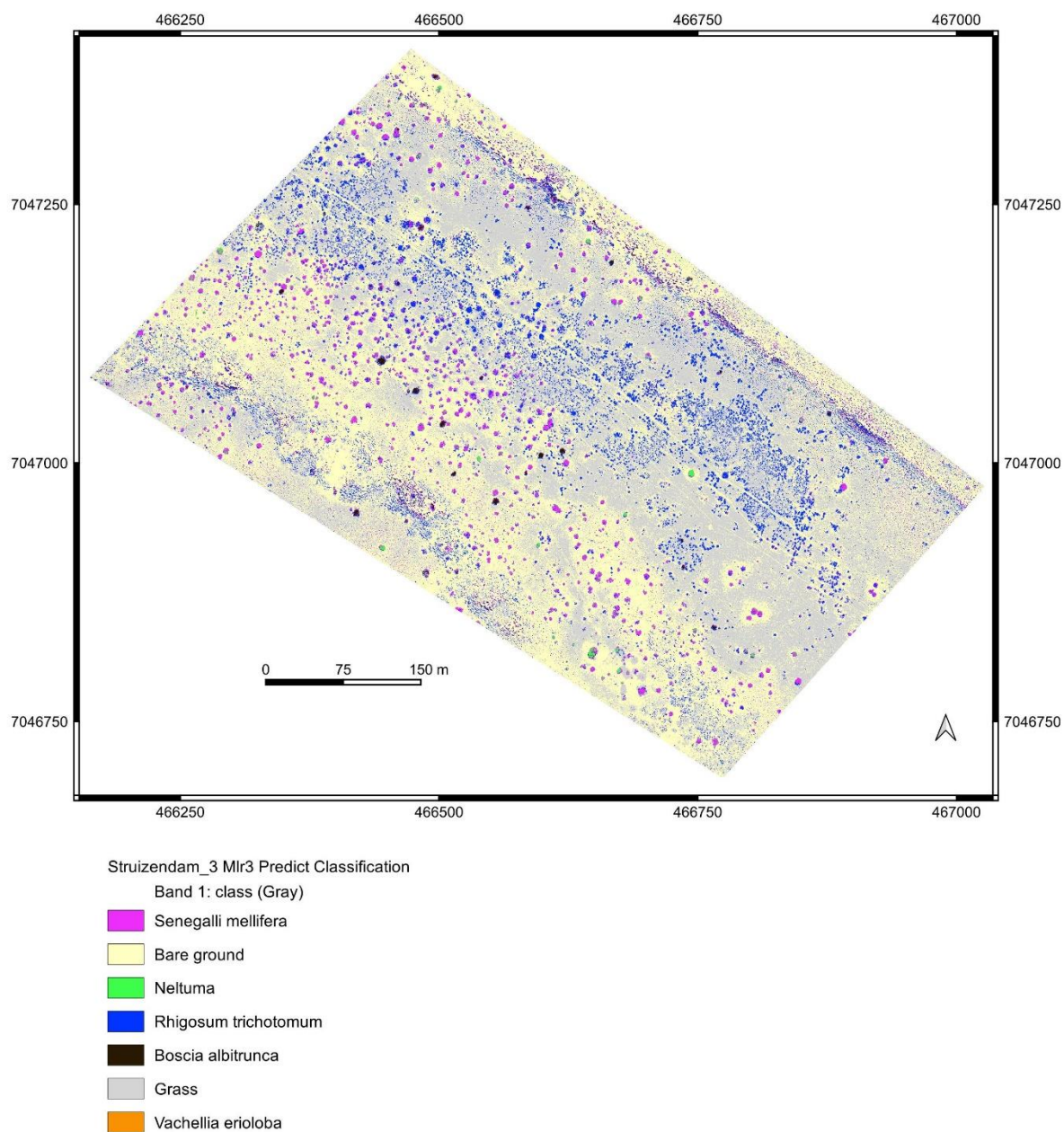


Figure S6. Final classification maps produced for the “Struizendam 3” drone survey area using the mlr3 pipeline. Best performing classification using SVM learner and Drone+CHM+AIIVI data set.

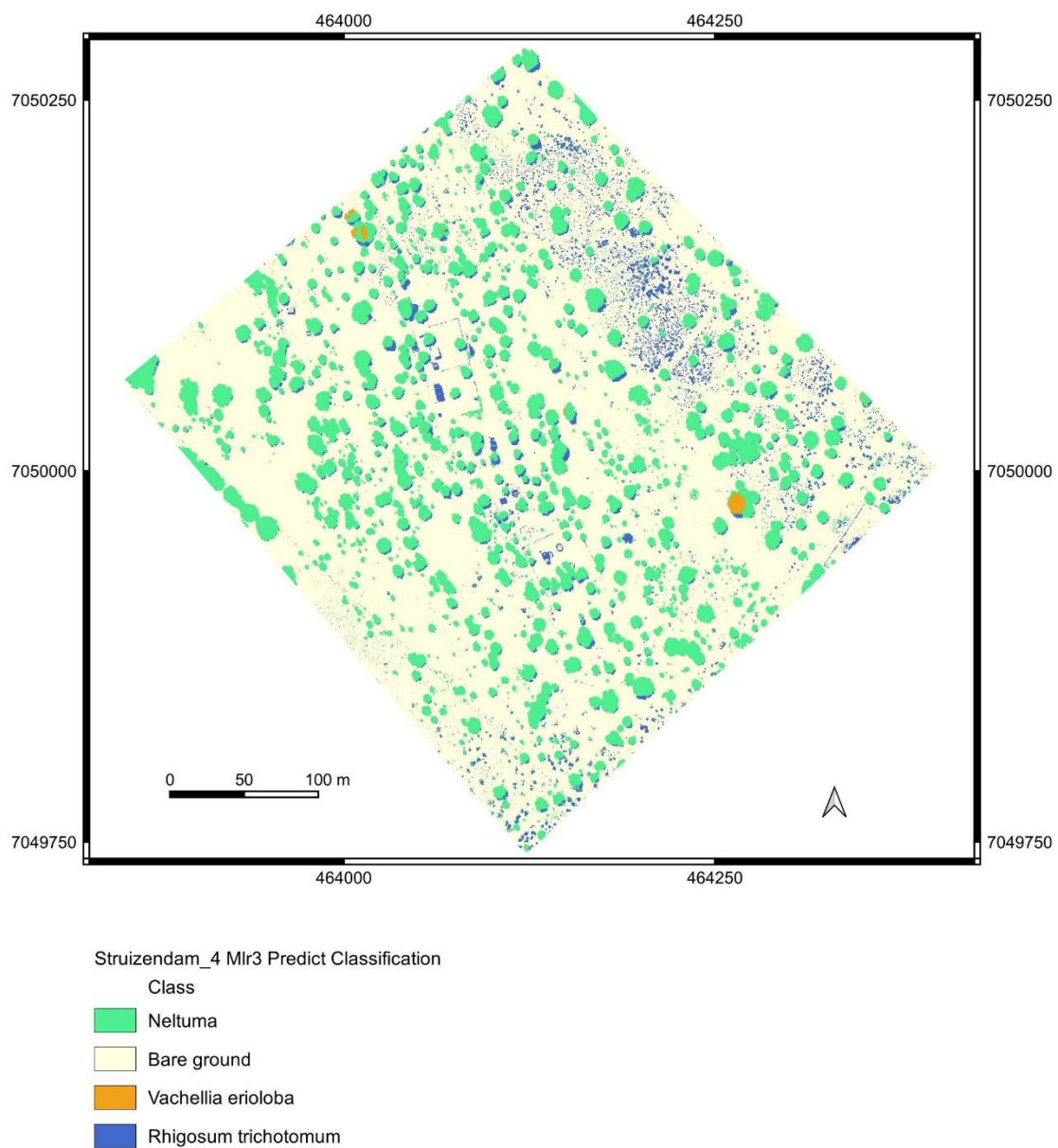


Figure S7. Final classification maps produced for the “Struizendam 4” drone survey area using the mlr3 pipeline. Best performing classification using SVM learner and Drone+CHM+AIIVI data set.

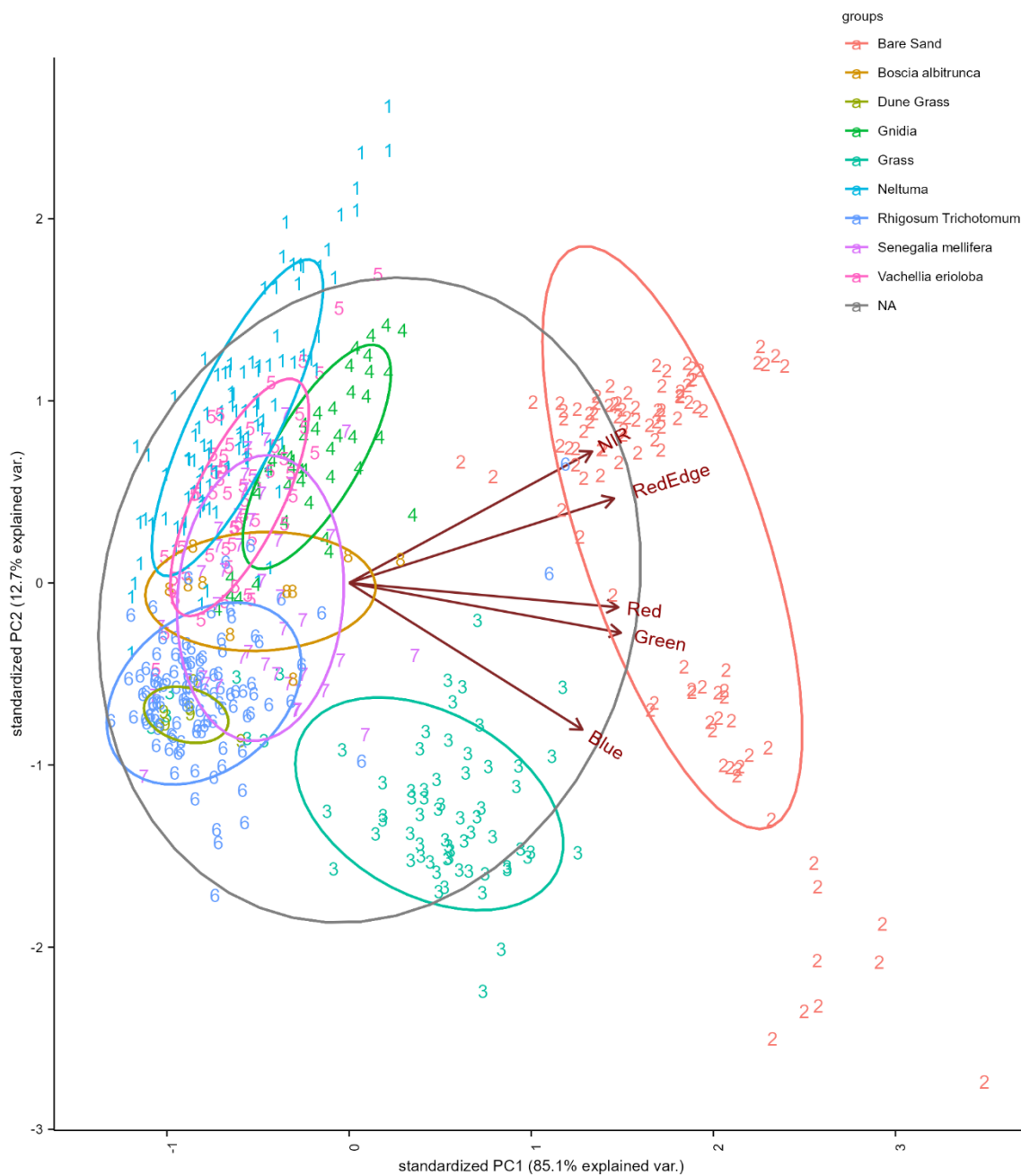


Figure S8. Principle Component Analysis of the spectral reflectance in each multispectral drone sensor band sample for main vegetation types.

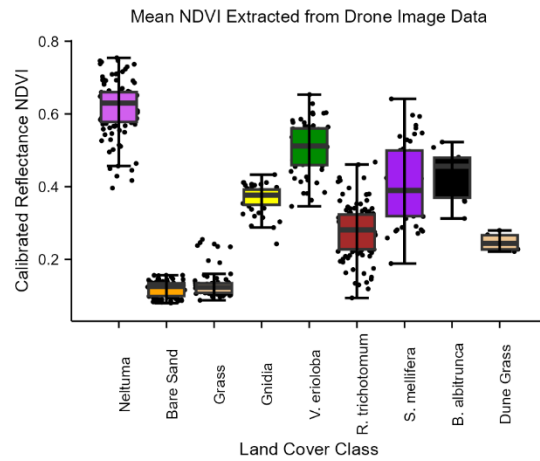
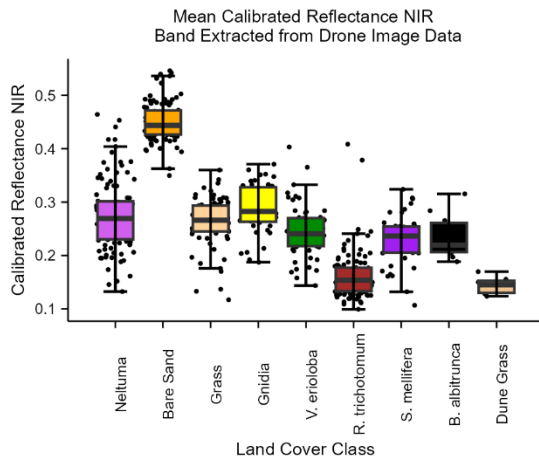
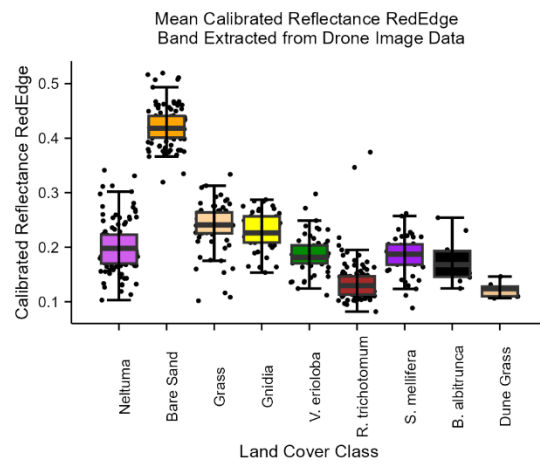
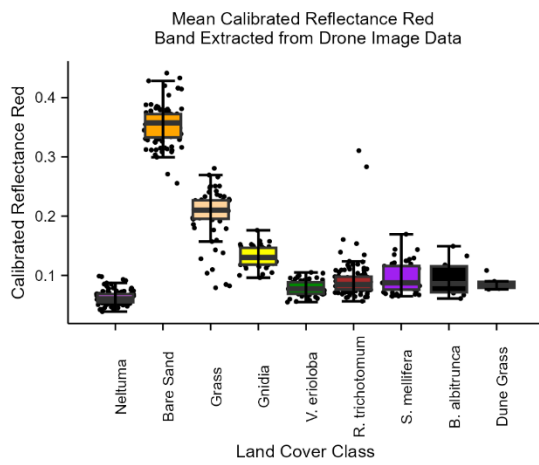
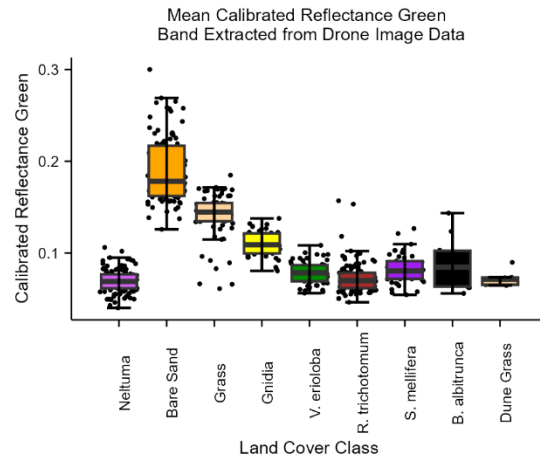
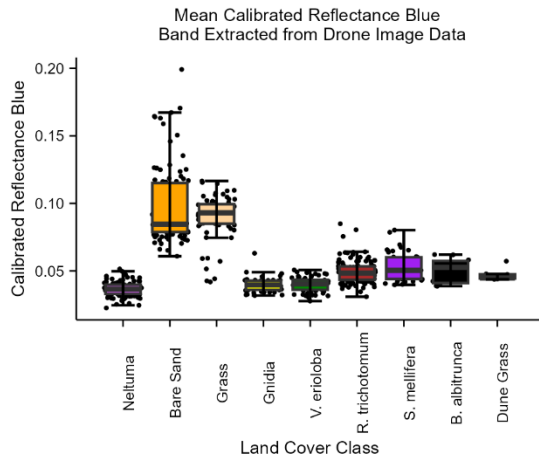


Figure S9. Box plots of spectral reflectance sampled for different vegetation types in each of the drone multispectral sensor bands and NDVI.

Changes in Vegetation Cover with distance from settlements

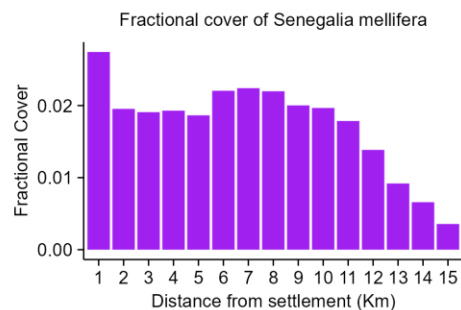
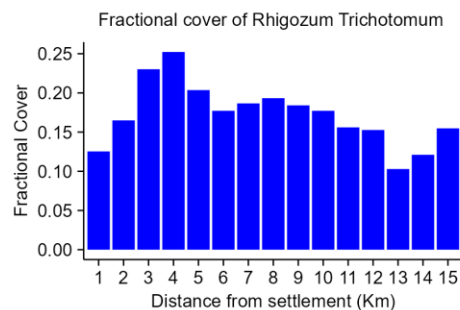
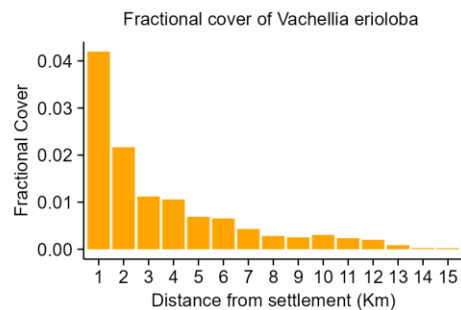
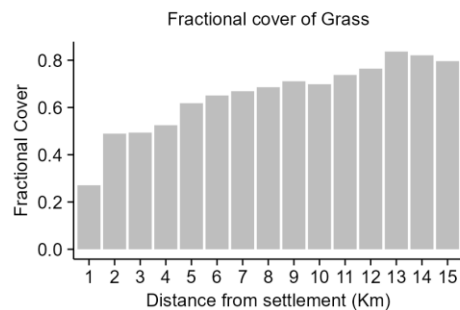
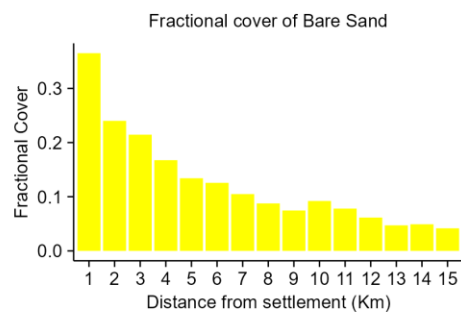
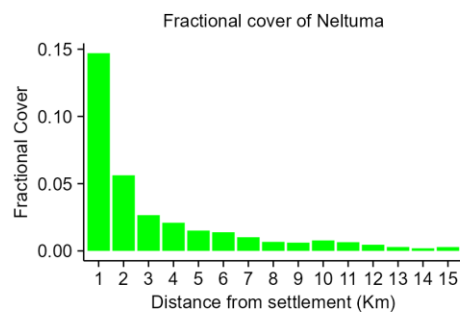
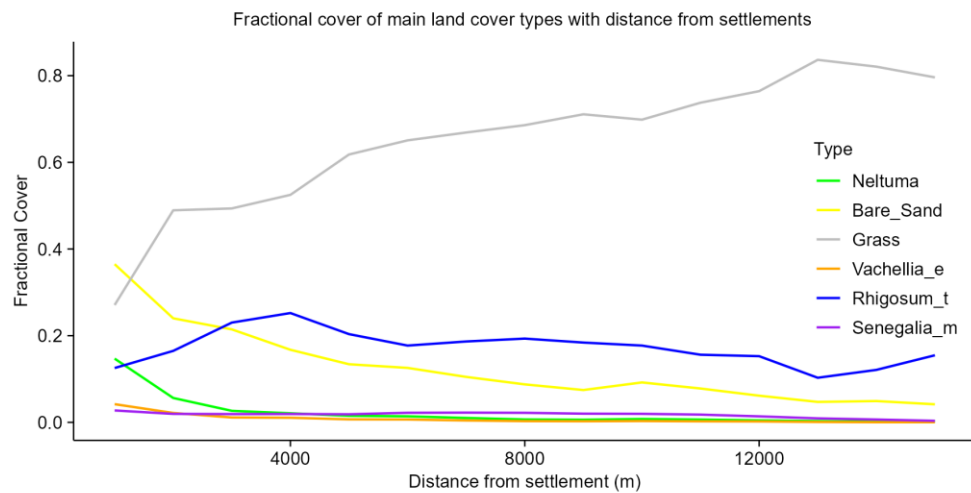


Figure S10. Fractional cover of the principal land-cover classes in relation to distance from settlements. These exploratory visualisations illustrate broad spatial associations between mapped vegetation cover and settlement proximity and are intended to support interpretation of landscape-scale invasion patterns rather than imply causal relationships

Changes in Vegetation Cover with distance from main roads

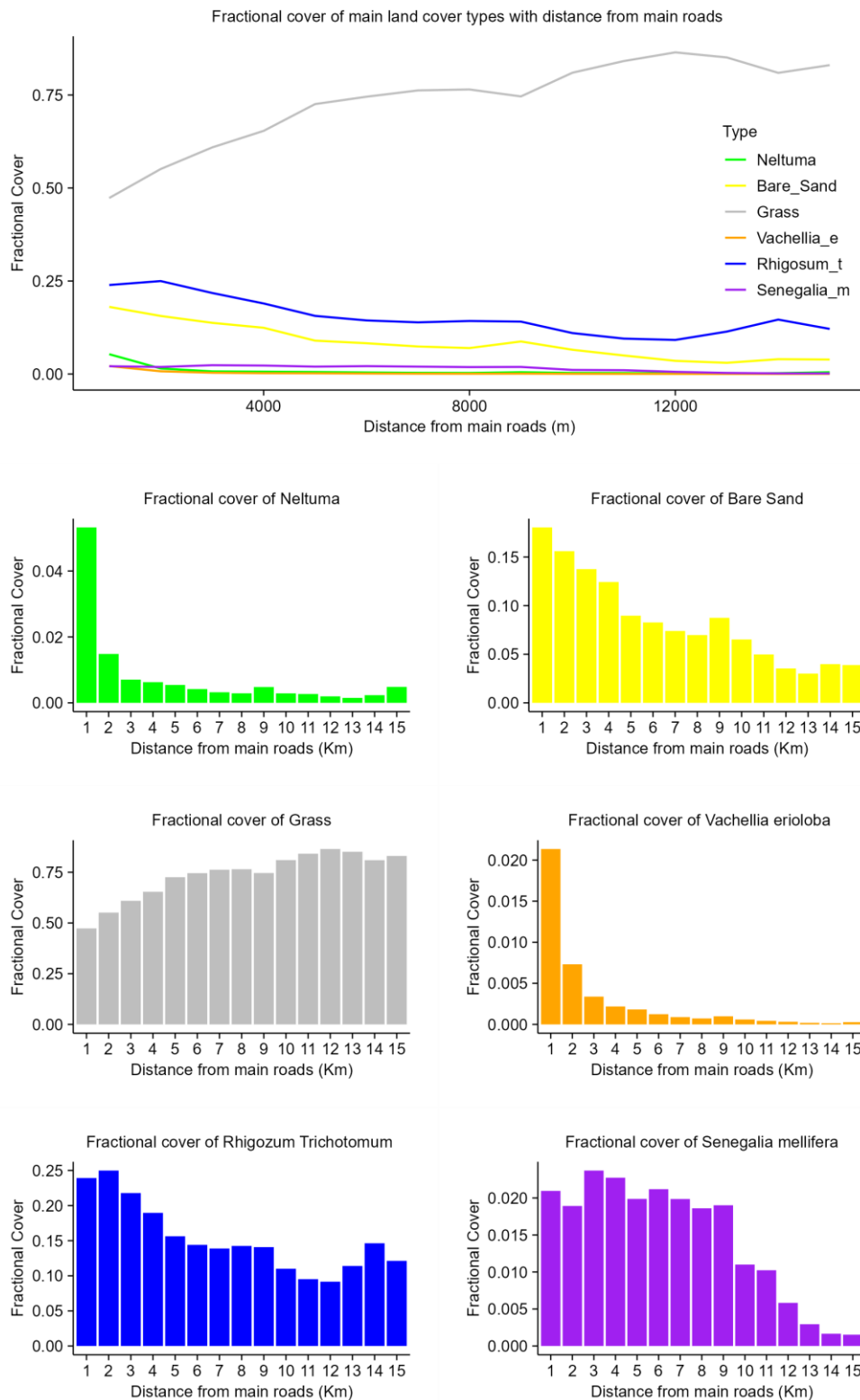


Figure S11. Fractional cover of the principal land-cover classes in relation to distance from main roads. In the study area, roads are predominantly aligned with ephemeral dry riverbeds, and these exploratory visualisations therefore reflect the combined spatial association of roads, settlements and river corridors. They are intended to support interpretation of landscape-scale invasion patterns rather than imply causal relationships.

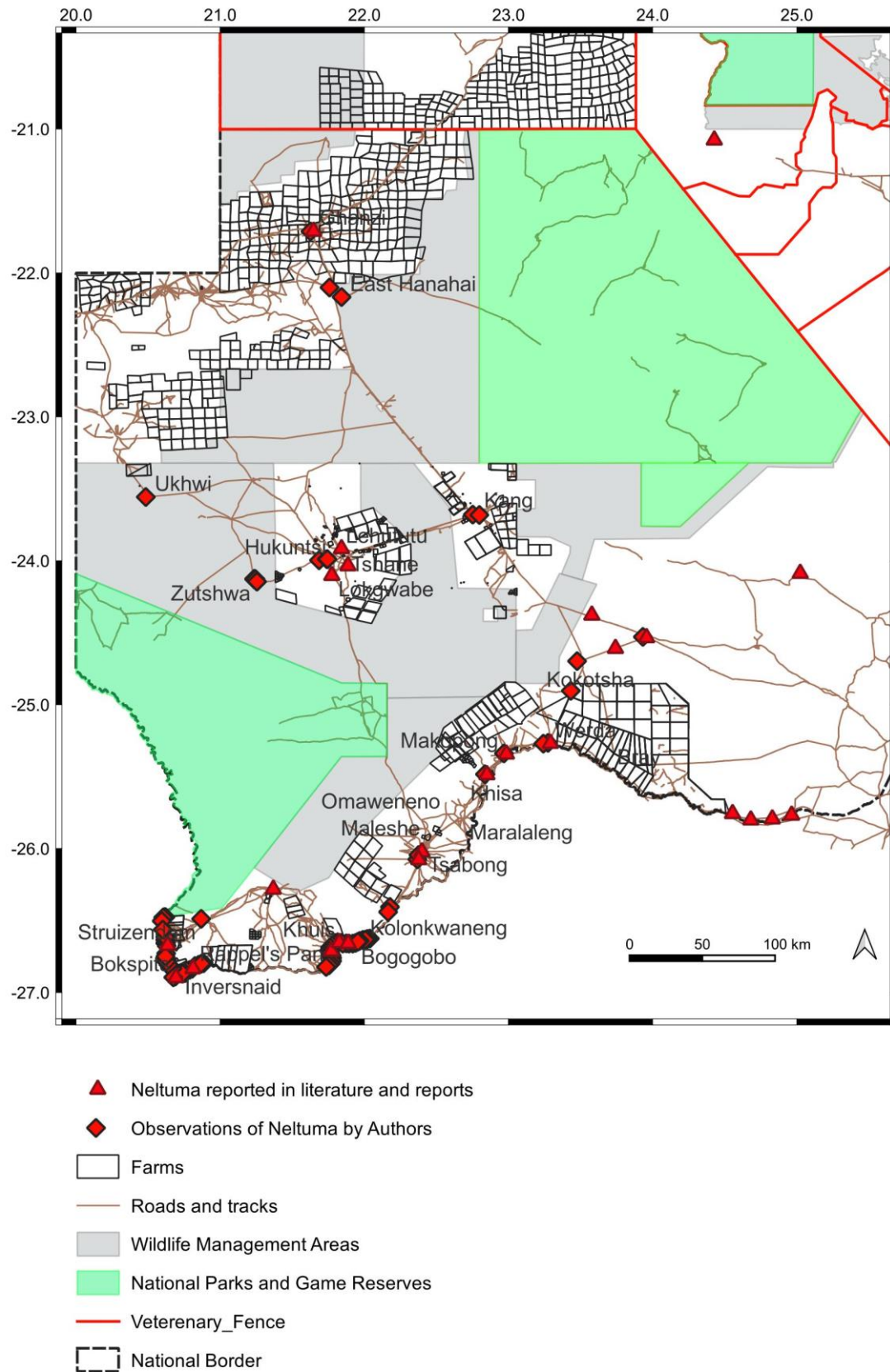


Figure S12. *Neltuma* occurrence in western Botswana. Personal field observations by the authors and locations reported in the literature (*Integrated Mesquite (Prosopis Species) Management Strategy for Botswana* 2022; Mosweu et al. 2013; Muzila et al. 2010).

References

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